PYTHON PROGRAMMING

OUTPUT SCREENSHOTS

Name: Thota Ganesh
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Course / Section: B.Tech, CSM-3

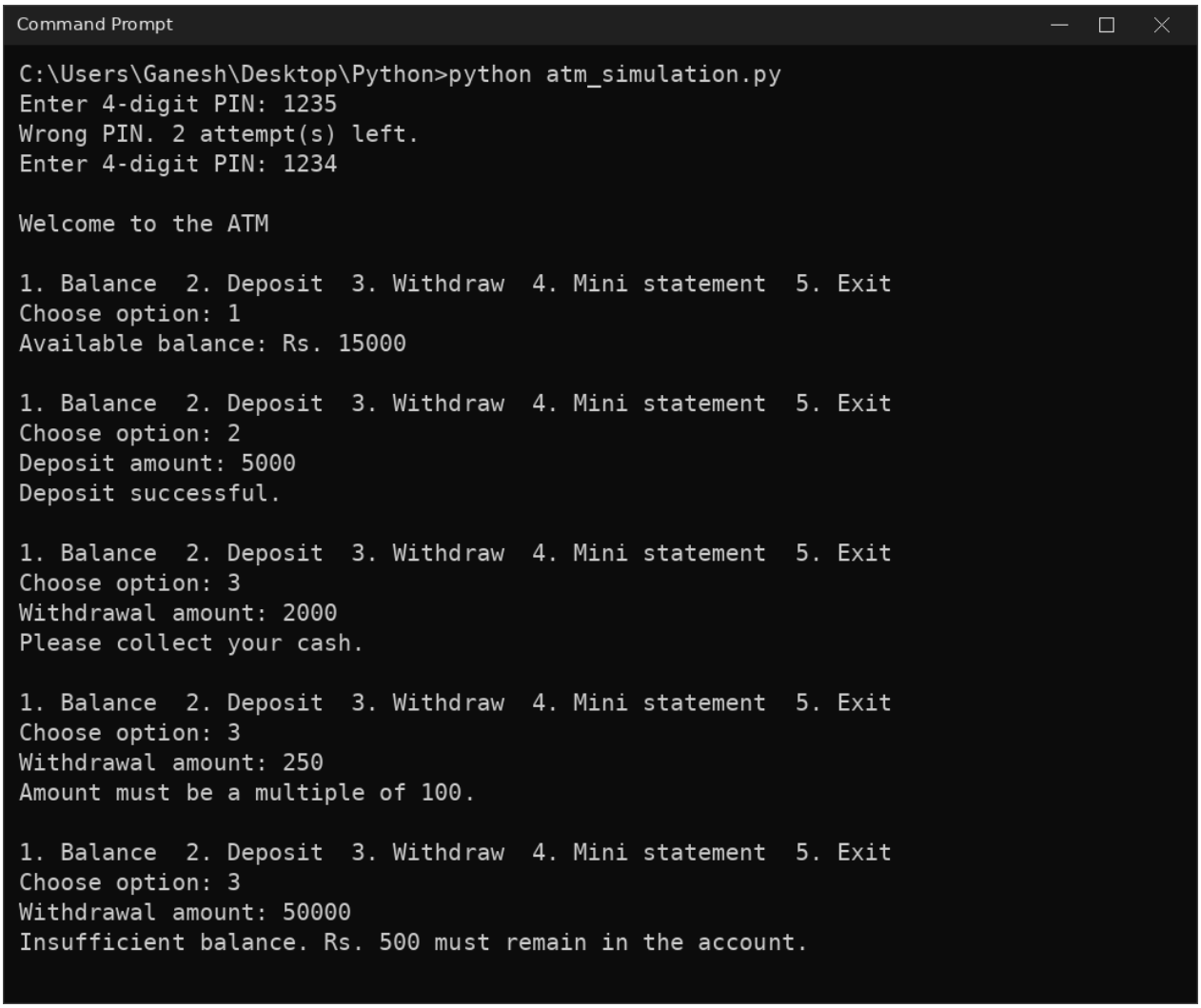
Index

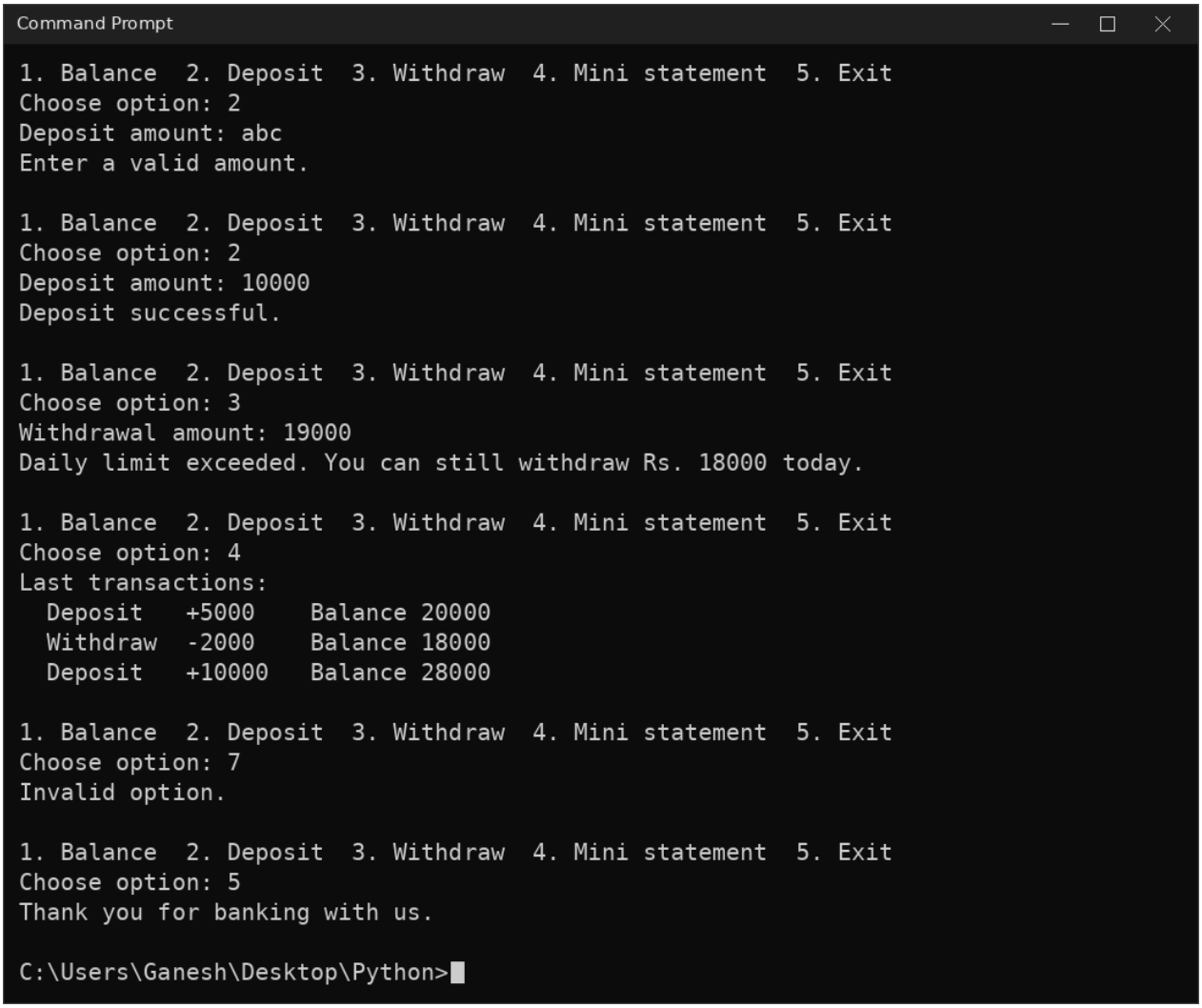
1. ATM Simulation
2. Attendance Recovery Planner
3. Campus Event Budget Planning
4. Employee Payroll
5. Exam Result with Grace Marks
6. Hospital Billing
7. Manufacturing Profit Analysis
8. Progressive Electricity Billing
9. Retail Billing
10. Travel Budget Planner

Program 1: ATM Simulation

File: atm_simulation.py

Output 1:




Command Prompt
1. Balance 2. Deposit 3. Withdraw 4. Mini statement 5. Exit
Choose option: 2
Deposit amount: abc
Enter a valid amount.
1. Balance 2. Deposit 3. Withdraw 4. Mini statement 5. Exit
Choose option: 2
Deposit amount: 10000
Deposit successful.
1. Balance 2. Deposit 3. Withdraw 4. Mini statement 5. Exit
Choose option: 3
Withdrawal amount: 19000
Daily limit exceeded. You can still withdraw Rs. 18000 today.
1. Balance 2. Deposit 3. Withdraw 4. Mini statement 5. Exit
Choose option: 4
Last transactions:
Deposit +5000 Balance 20000
Withdraw -2000 Balance 18000
Deposit +10000 Balance 28000
1. Balance 2. Deposit 3. Withdraw 4. Mini statement 5. Exit
Choose option: 7
Invalid option.
1. Balance 2. Deposit 3. Withdraw 4. Mini statement 5. Exit
Choose option: 5
Thank you for banking with us.
C:\Users\Ganesh\Desktop\Python>

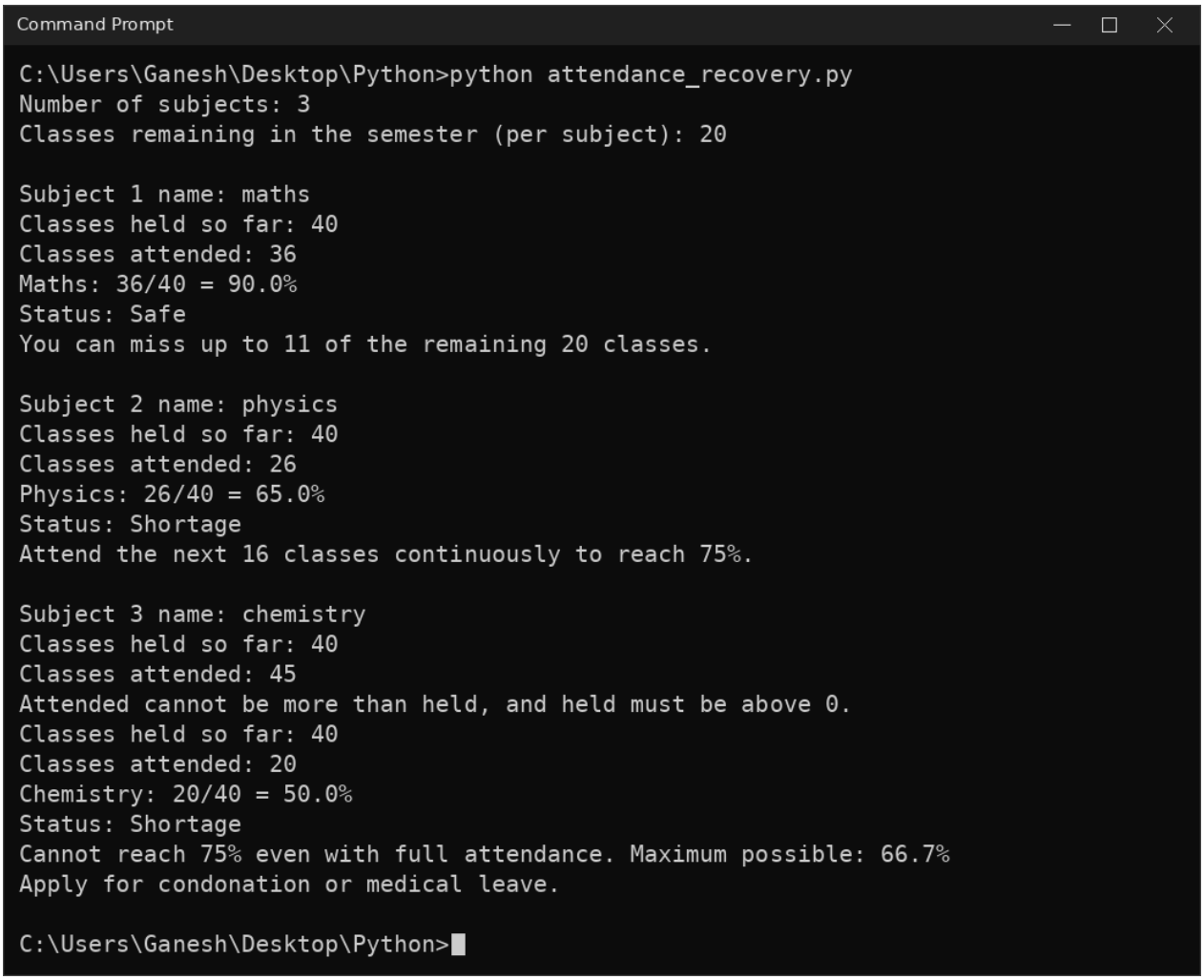
Output 2:


Command Prompt
C:\Users\Ganesh\Desktop\Python>python atm_simulation.py
Enter 4-digit PIN: 1111
Wrong PIN. 2 attempt(s) left.
Enter 4-digit PIN: 2222
Wrong PIN. 1 attempt(s) left.
Enter 4-digit PIN: 3333
Wrong PIN. 0 attempt(s) left.
Card blocked. Please contact your bank.
C:\Users\Ganesh\Desktop\Python>

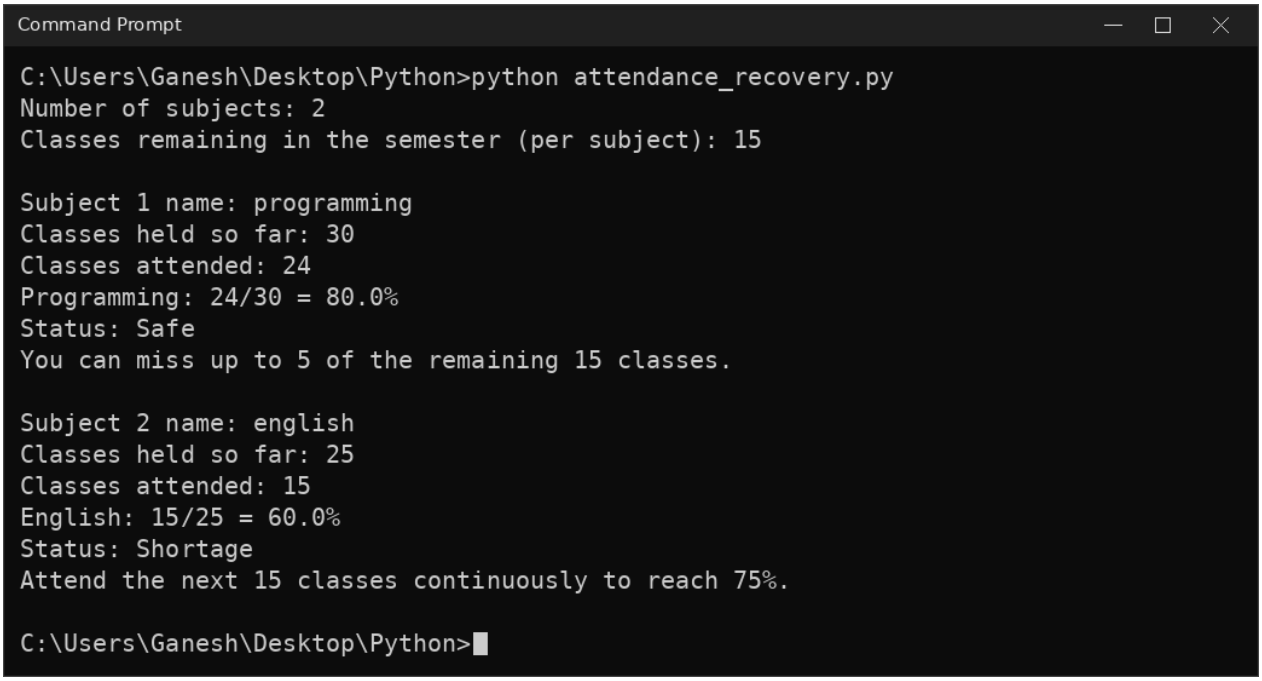
Program 2: Attendance Recovery Planner

File: attendance_recovery.py

Output 1:



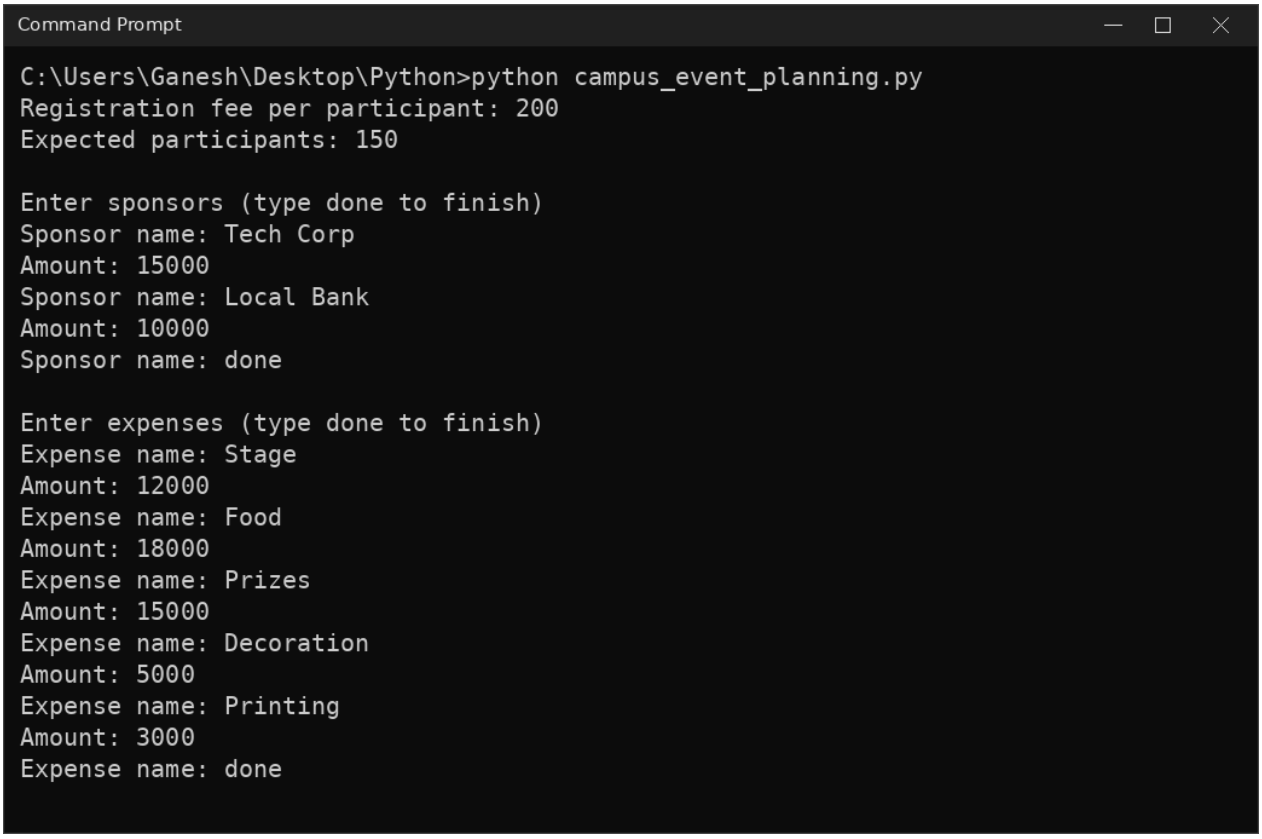
Output 2:



Program 3: Campus Event Budget Planning

File: campus_event_planning.py

Output 1:



```
Command Prompt
-----------------------------------------
            EVENT FINANCIAL PLAN
-----------------------------------------
Registration fees                   30000.00
Sponsor: Tech Corp                  15000.00
Sponsor: Local Bank                 10000.00
Total income                        55000.00
-----------------------------------------
Stage                    12000.00    22.6%  Major
Food                     18000.00    34.0%  Major
Prizes                   15000.00    28.3%  Major
Decoration                5000.00     9.4%
Printing                  3000.00     5.7%
Total expenses                      53000.00
-----------------------------------------
Surplus : Rs. 2000.00
Status  : Risky - safety margin below 10%
Break-even participants: 140

C:\Users\Ganesh\Desktop\Python>
```

Output 2:

```
Command Prompt
C:\Users\Ganesh\Desktop\Python>python campus_event_planning.py
Registration fee per participant: 100
Expected participants: 80

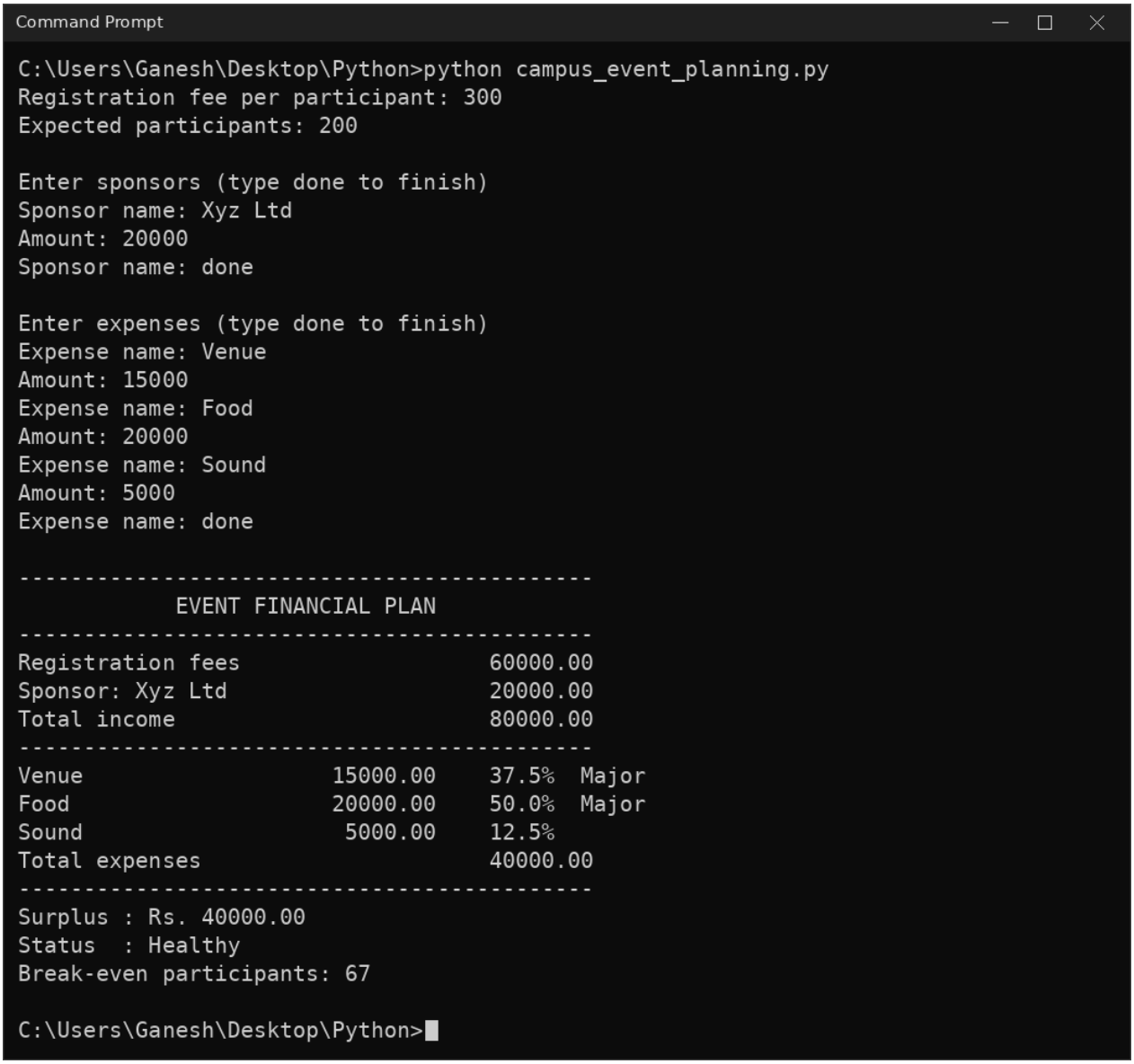
Enter sponsors (type done to finish)
Sponsor name: done

Enter expenses (type done to finish)
Expense name: Venue
Amount: 10000
Expense name: Food
Amount: 6000
Expense name: done

-----------------------------------------
            EVENT FINANCIAL PLAN
-----------------------------------------
Registration fees                    8000.00
Total income                         8000.00
-----------------------------------------
Venue                    10000.00    62.5%  Major
Food                      6000.00    37.5%  Major
Total expenses                      16000.00
-----------------------------------------
Deficit : Rs. 8000.00
Status  : Deficit - reduce expenses or raise the fee
Break-even participants: 160

C:\Users\Ganesh\Desktop\Python>
```

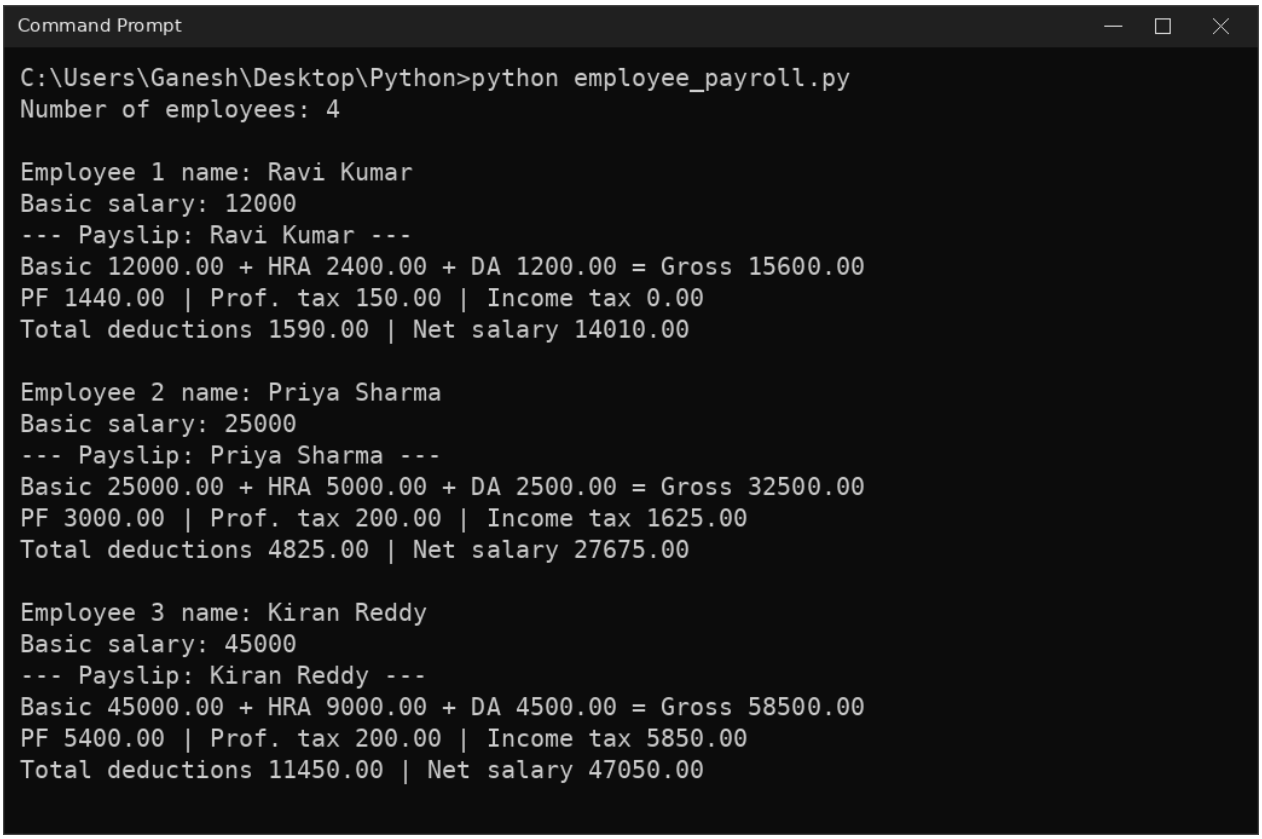
Output 3:

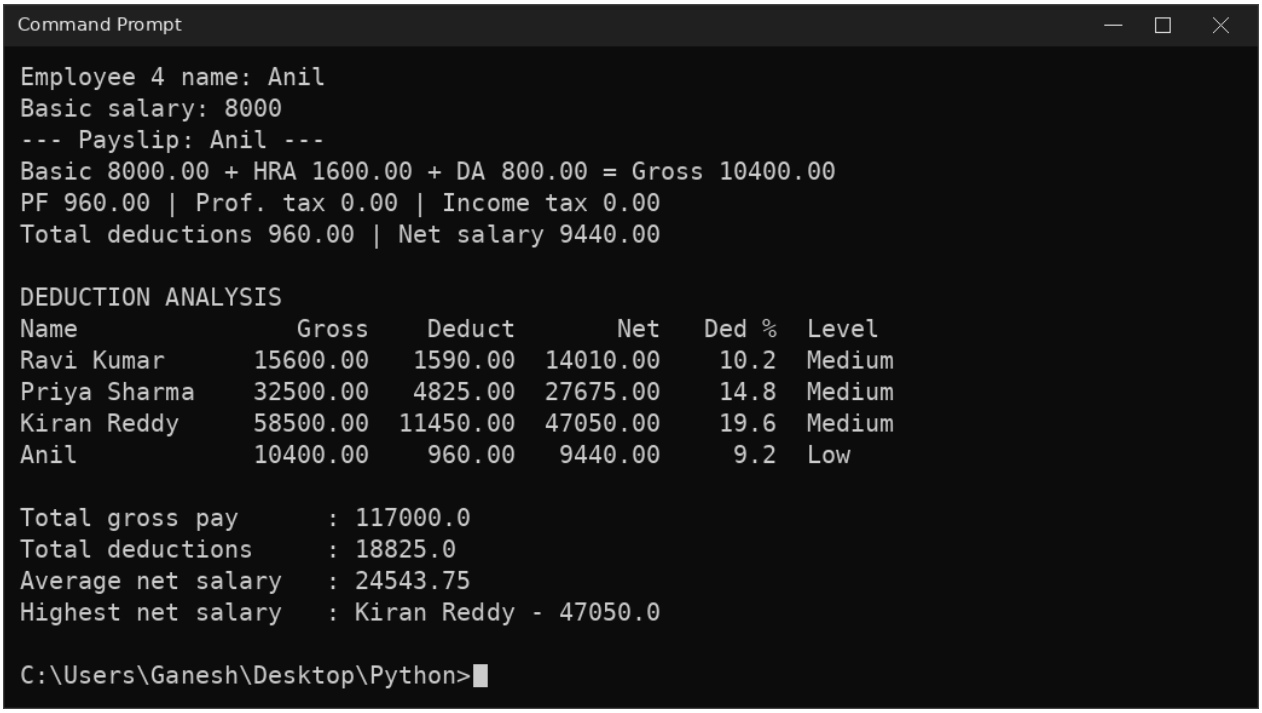

Command Prompt
C:\Users\Ganesh\Desktop\Python>python campus_event_planning.py
Registration fee per participant: 300
Expected participants: 200
Enter sponsors (type done to finish)
Sponsor name: Xyz Ltd
Amount: 20000
Sponsor name: done
Enter expenses (type done to finish)
Expense name: Venue
Amount: 15000
Expense name: Food
Amount: 20000
Expense name: Sound
Amount: 5000
Expense name: done
EVENT FINANCIAL PLAN
Registration fees 60000.00
Sponsor: Xyz Ltd 20000.00
Total income 80000.00
Venue 15000.00 37.5% Major
Food 20000.00 50.0% Major
Sound 5000.00 12.5%
Total expenses 40000.00
Surplus : Rs. 40000.00
Status : Healthy
Break-even participants: 67
C:\Users\Ganesh\Desktop\Python>

Program 4: Employee Payroll

File: employee_payroll.py

Output:

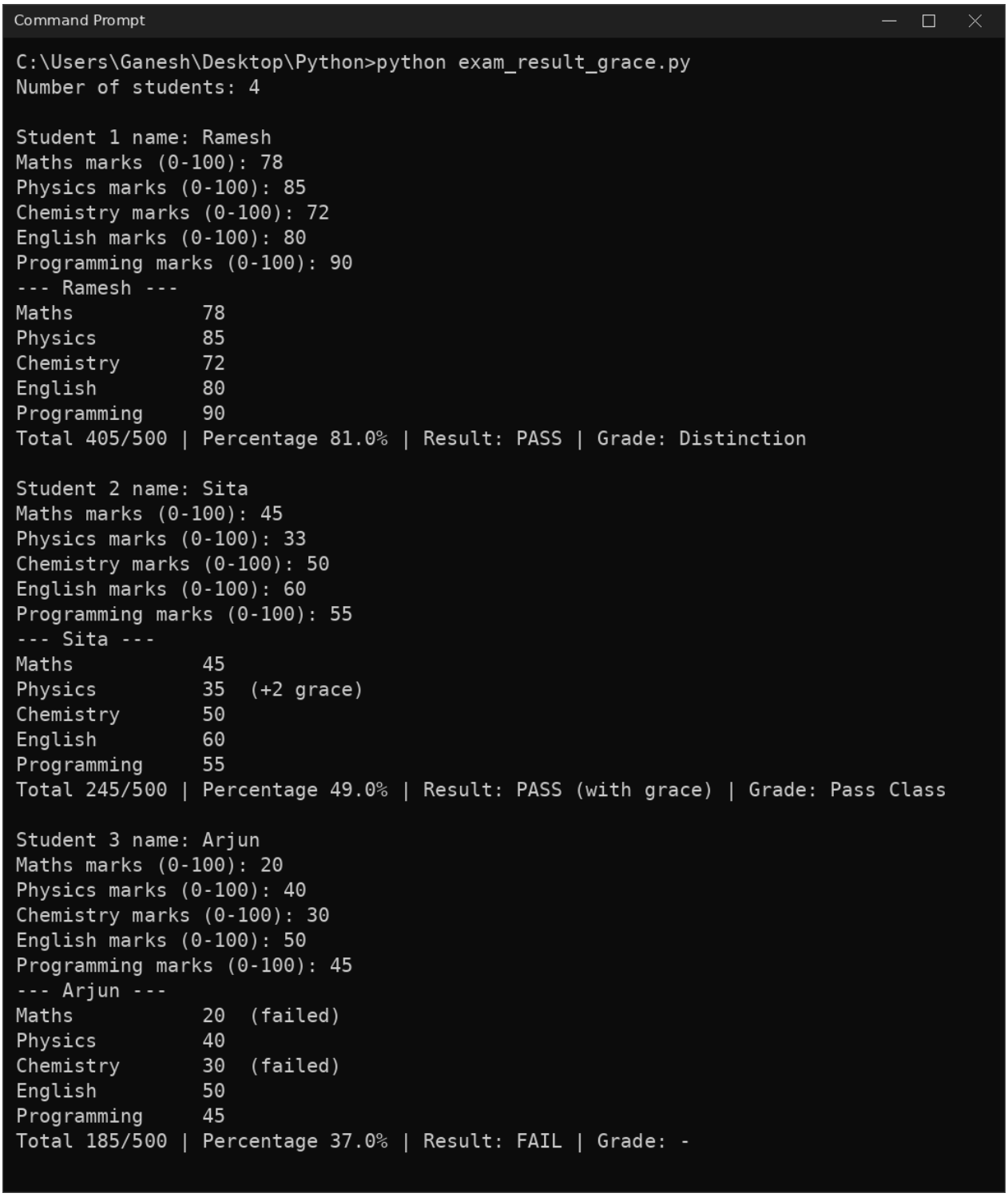


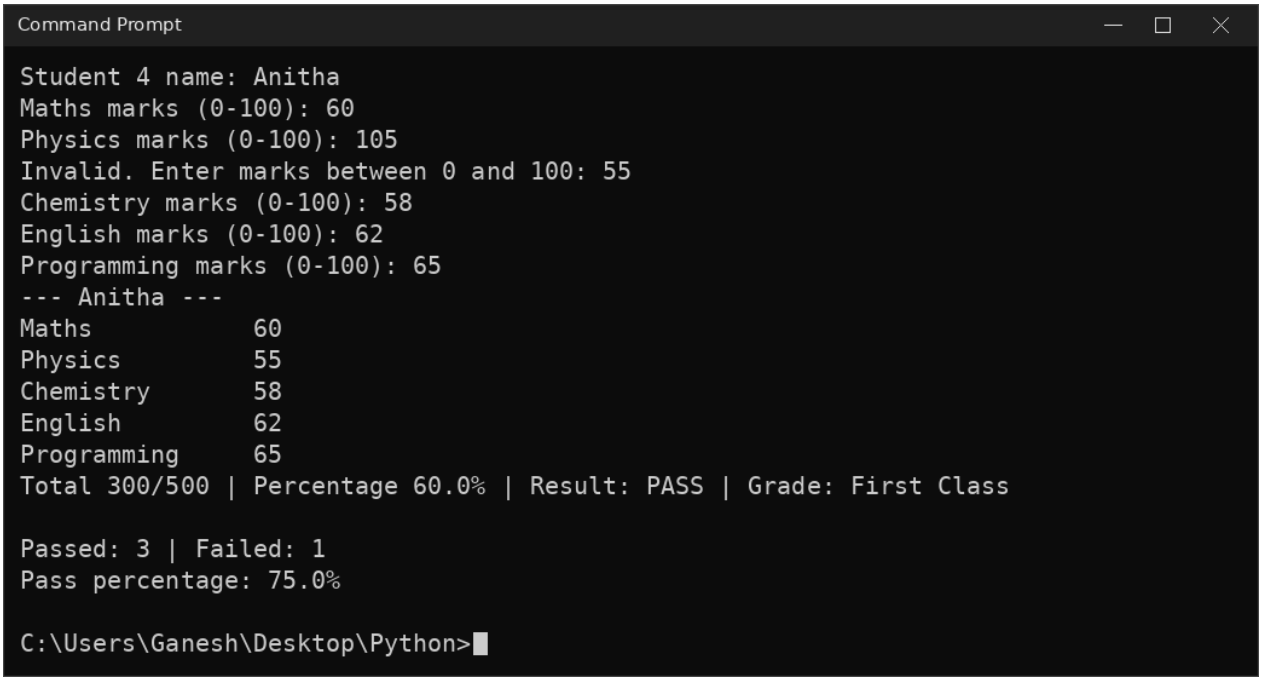


Program 5: Exam Result with Grace Marks

File: exam_result_grace.py

Output:

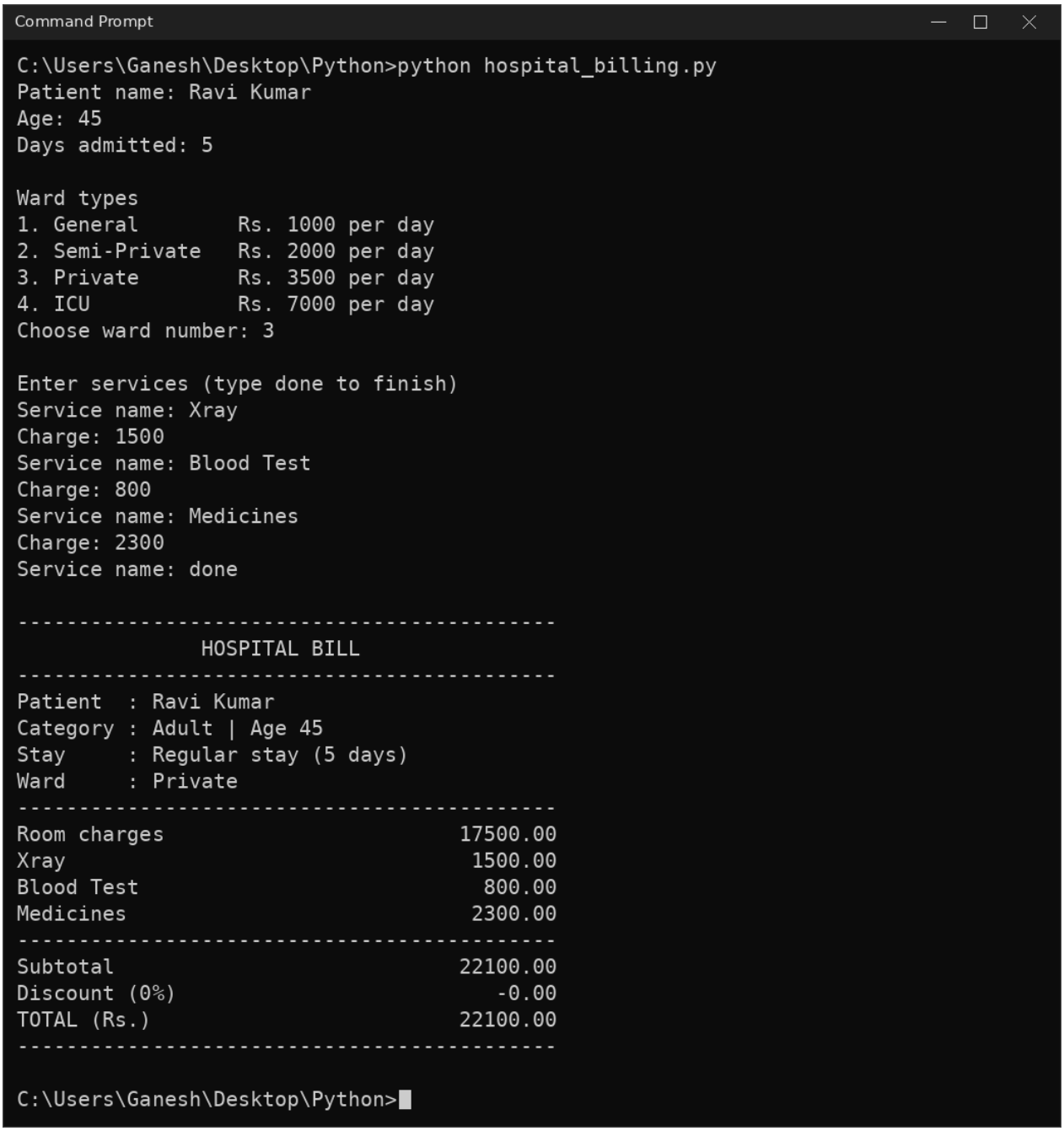



Command Prompt
Student 4 name: Anitha
Maths marks (0-100): 60
Physics marks (0-100): 105
Invalid. Enter marks between 0 and 100: 55
Chemistry marks (0-100): 58
English marks (0-100): 62
Programming marks (0-100): 65
--- Anitha ---
Maths 60
Physics 55
Chemistry 58
English 62
Programming 65
Total 300/500 | Percentage 60.0% | Result: PASS | Grade: First Class
Passed: 3 | Failed: 1
Pass percentage: 75.0%
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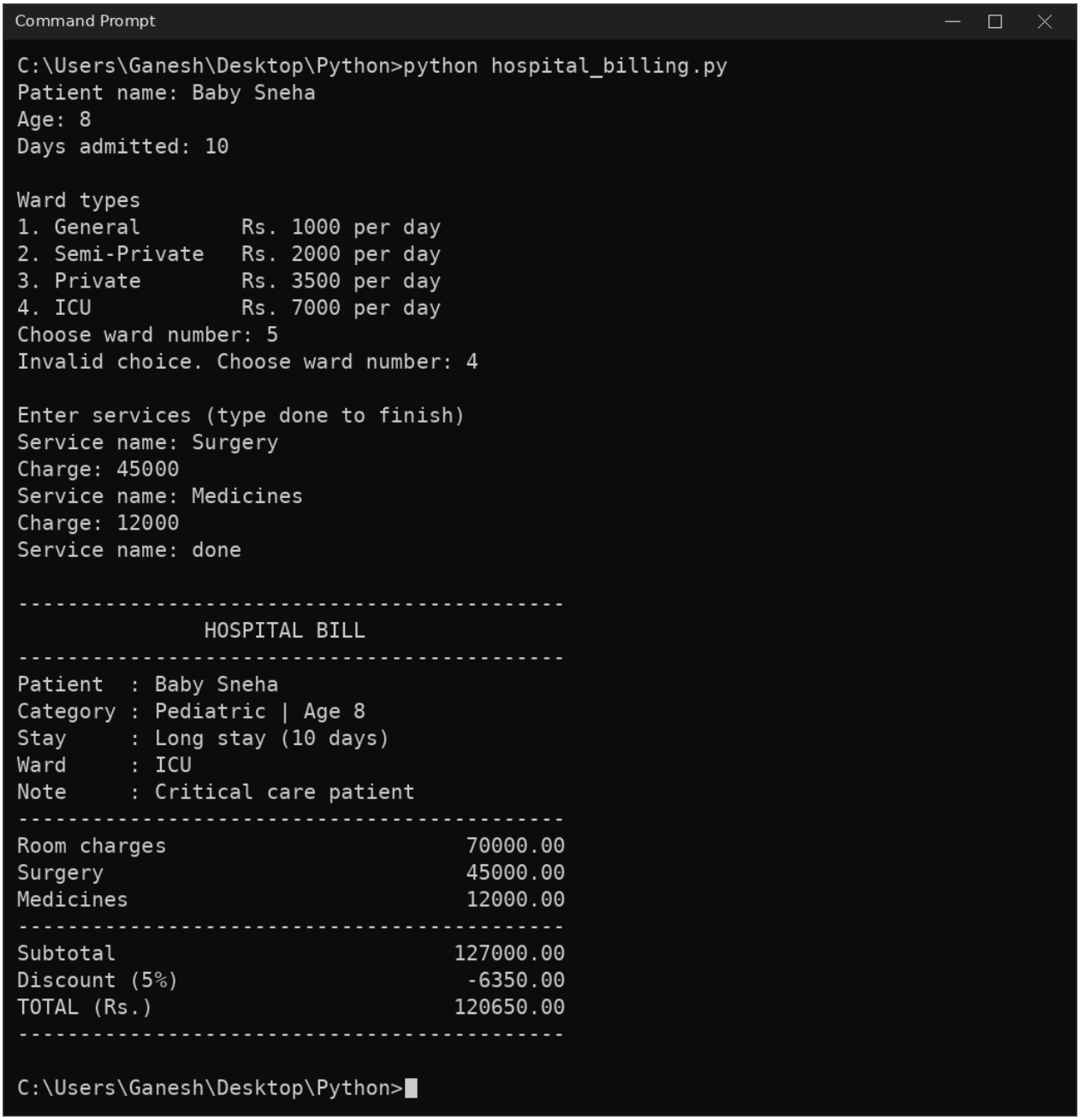
Program 6: Hospital Billing

File: hospital_billing.py

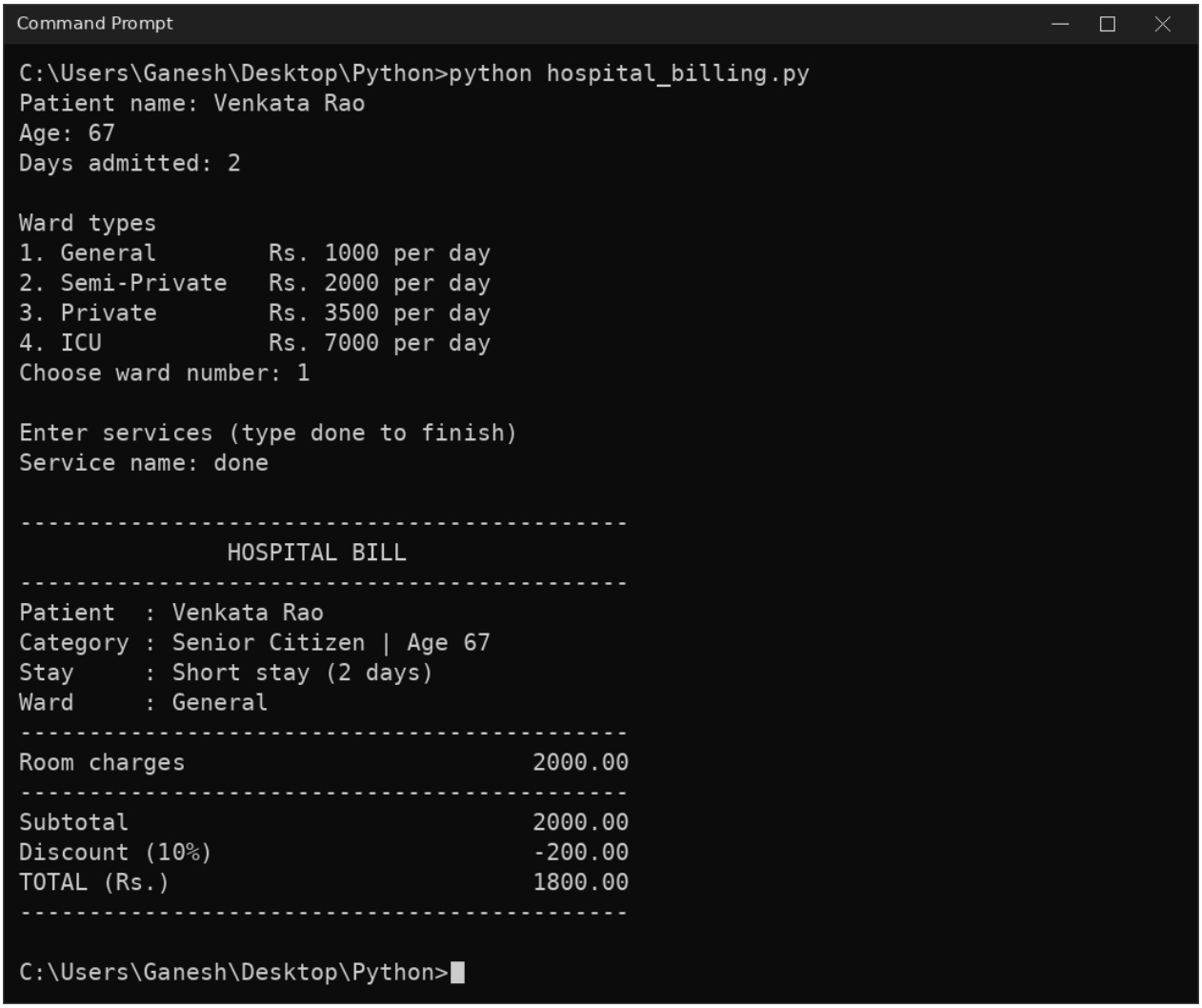
Output 1:



Output 2:


Command Prompt
C:\Users\Ganesh\Desktop\Python>python hospital_billing.py
Patient name: Baby Sneha
Age: 8
Days admitted: 10
Ward types
1. General Rs. 1000 per day
2. Semi-Private Rs. 2000 per day
3. Private Rs. 3500 per day
4. ICU Rs. 7000 per day
Choose ward number: 5
Invalid choice. Choose ward number: 4
Enter services (type done to finish)
Service name: Surgery
Charge: 45000
Service name: Medicines
Charge: 12000
Service name: done
HOSPITAL BILL
Patient : Baby Sneha
Category : Pediatric | Age 8
Stay : Long stay (10 days)
Ward : ICU
Note : Critical care patient
Room charges 70000.00
Surgery 45000.00
Medicines 12000.00
Subtotal 127000.00
Discount (5%) -6350.00
TOTAL (Rs.) 120650.00
C:\Users\Ganesh\Desktop\Python>

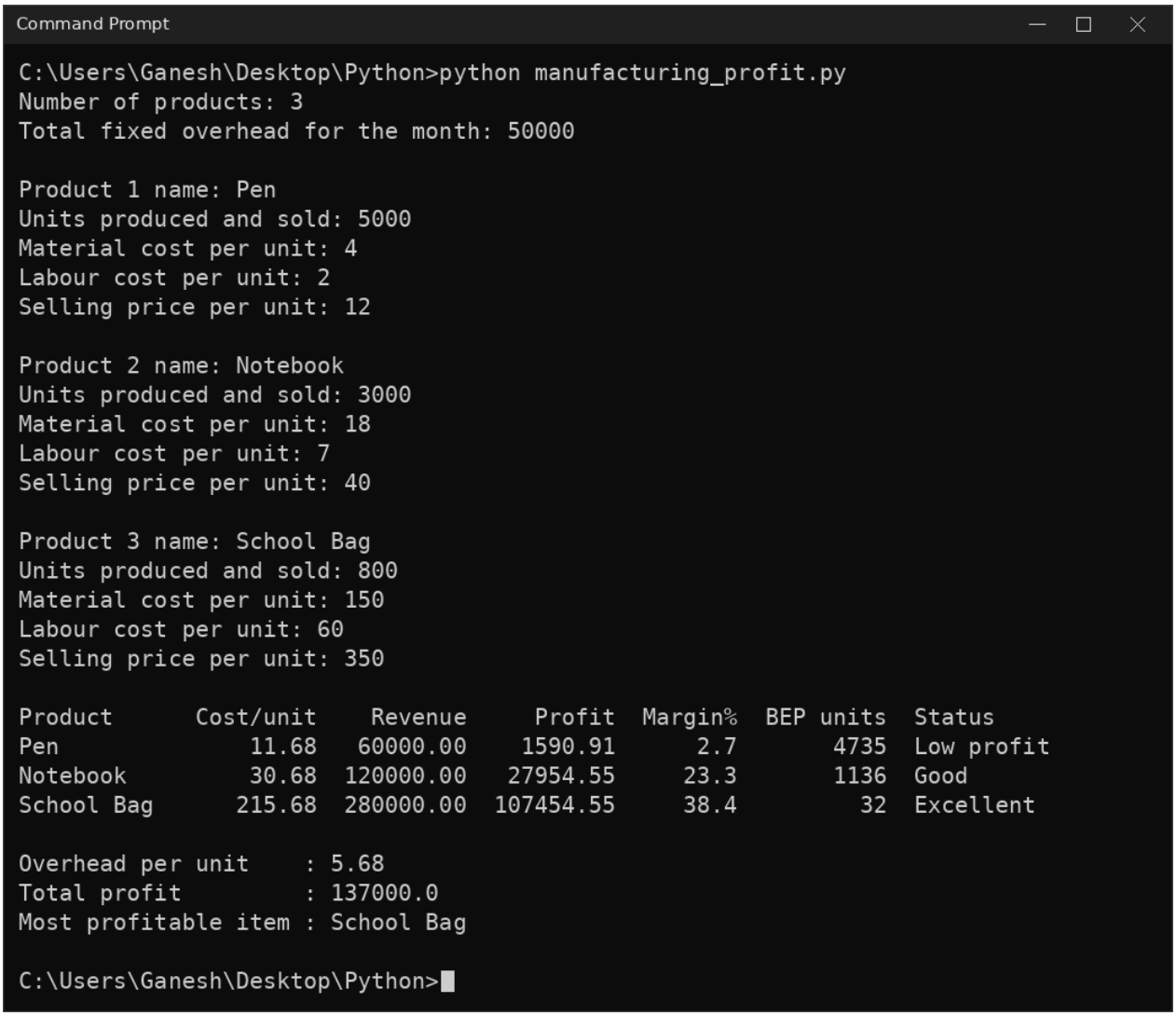
Output 3:


Command Prompt
C:\Users\Ganesh\Desktop\Python>python hospital_billing.py
Patient name: Venkata Rao
Age: 67
Days admitted: 2
Ward types
1. General Rs. 1000 per day
2. Semi-Private Rs. 2000 per day
3. Private Rs. 3500 per day
4. ICU Rs. 7000 per day
Choose ward number: 1
Enter services (type done to finish)
Service name: done
HOSPITAL BILL
Patient : Venkata Rao
Category : Senior Citizen | Age 67
Stay : Short stay (2 days)
Ward : General
Room charges 2000.00
Subtotal 2000.00
Discount (10%) -200.00
TOTAL (Rs.) 1800.00
C:\Users\Ganesh\Desktop\Python>

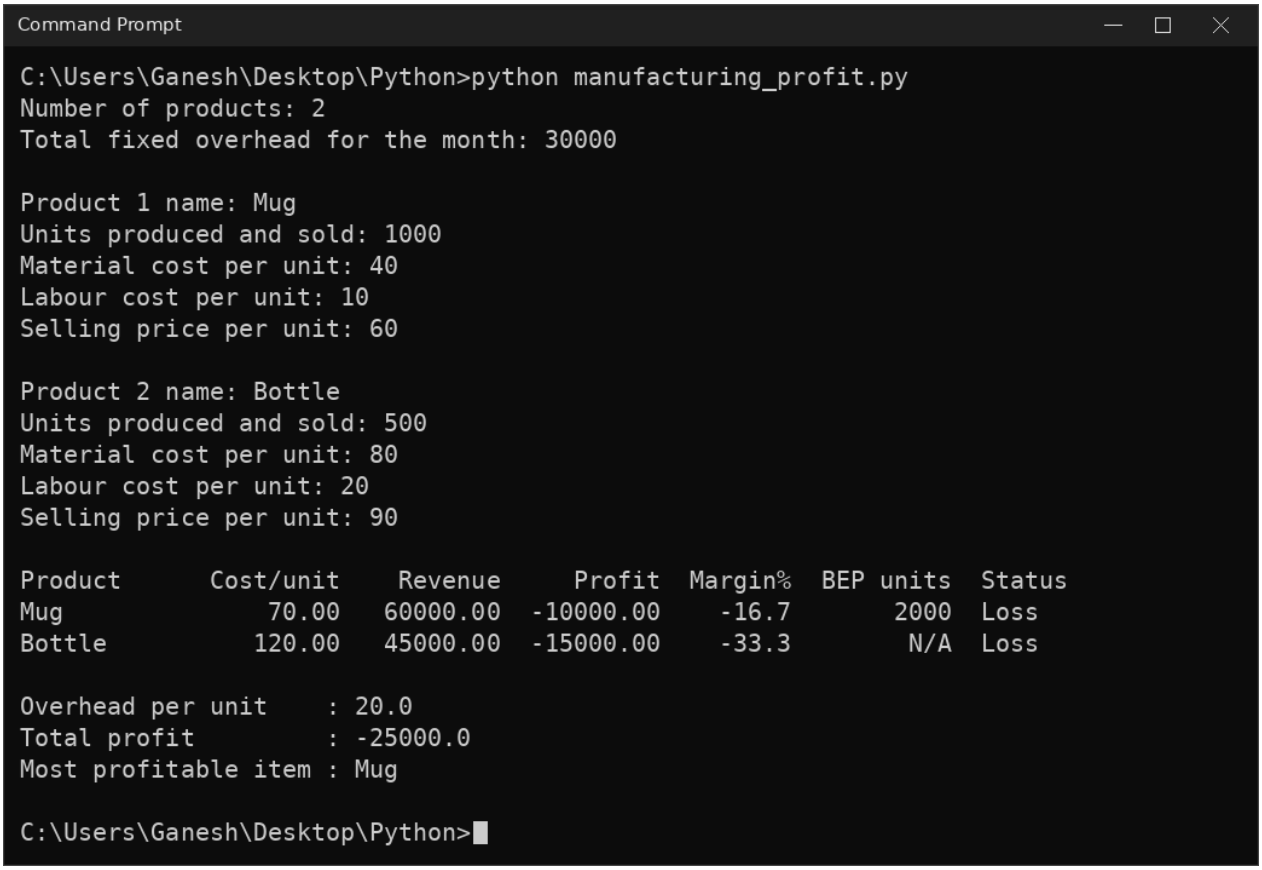
Program 7: Manufacturing Profit Analysis

File: manufacturing_profit.py

Output 1:



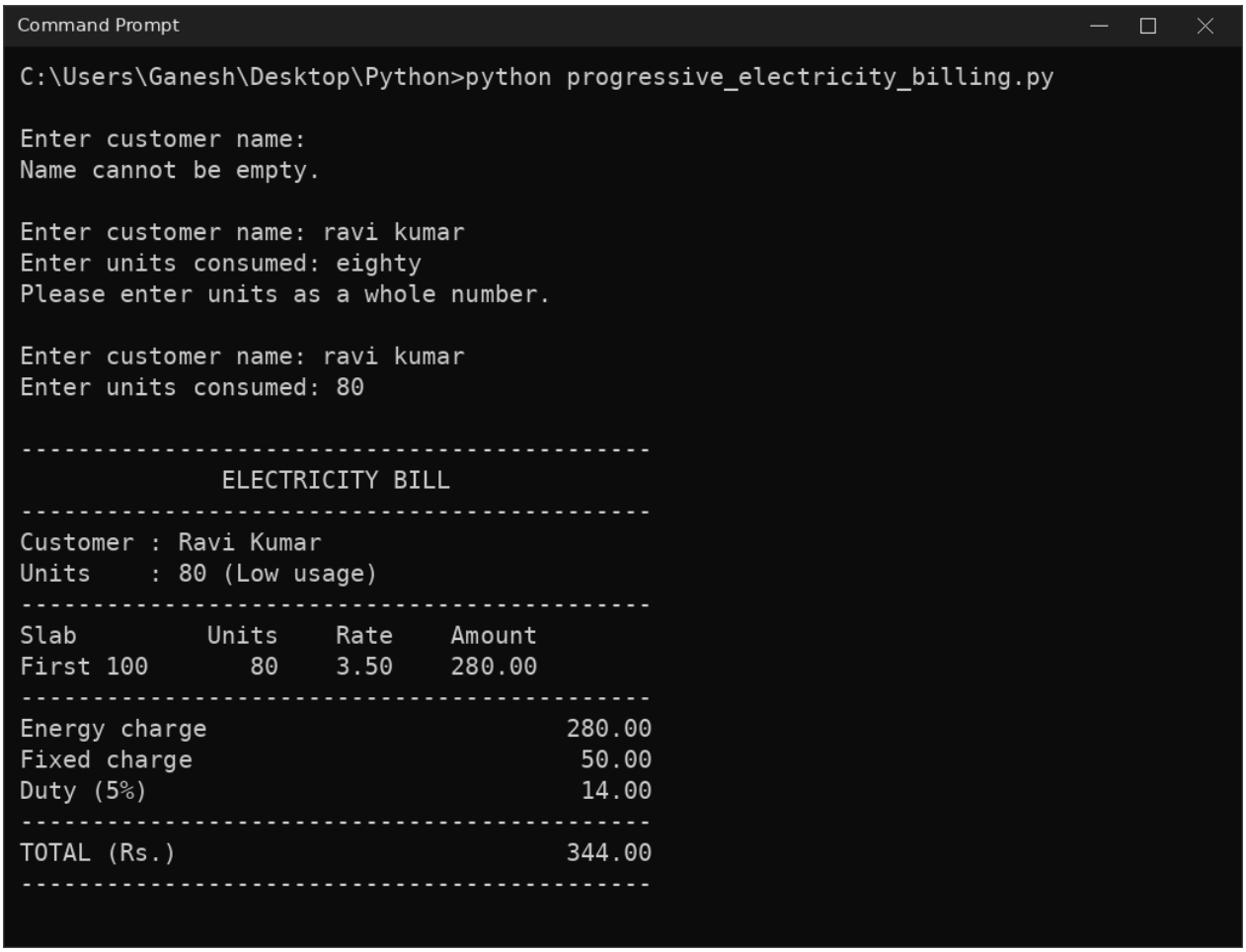
Output 2:

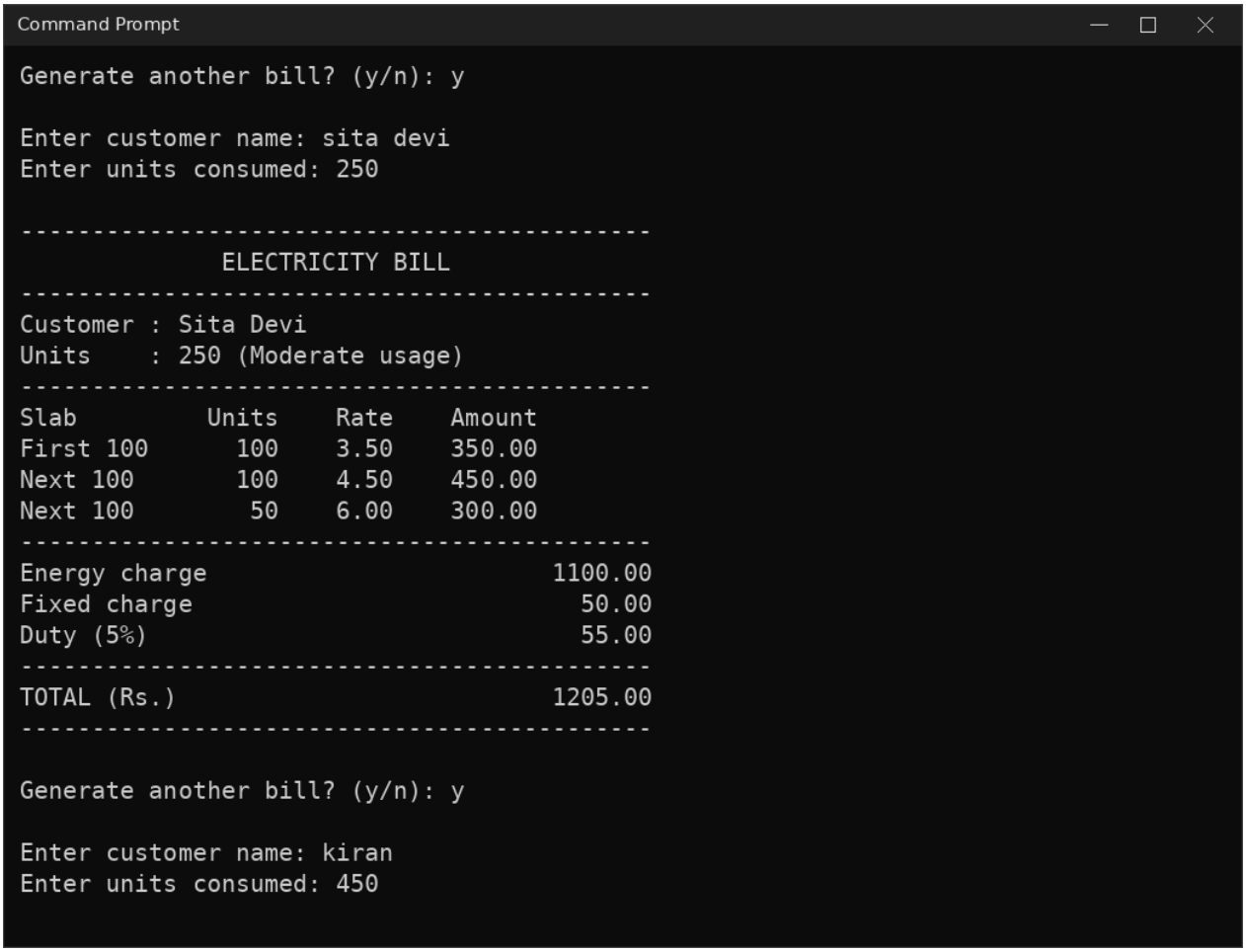

Command Prompt
C:\Users\Ganesh\Desktop\Python>python manufacturing_profit.py
Number of products: 2
Total fixed overhead for the month: 30000
Product 1 name: Mug
Units produced and sold: 1000
Material cost per unit: 40
Labour cost per unit: 10
Selling price per unit: 60
Product 2 name: Bottle
Units produced and sold: 500
Material cost per unit: 80
Labour cost per unit: 20
Selling price per unit: 90
Product Cost/unit Revenue Profit Margin% BEP units Status
Mug 70.00 60000.00 -10000.00 -16.7 2000 Loss
Bottle 120.00 45000.00 -15000.00 -33.3 N/A Loss
Overhead per unit : 20.0
Total profit : -25000.0
Most profitable item : Mug
C:\Users\Ganesh\Desktop\Python>

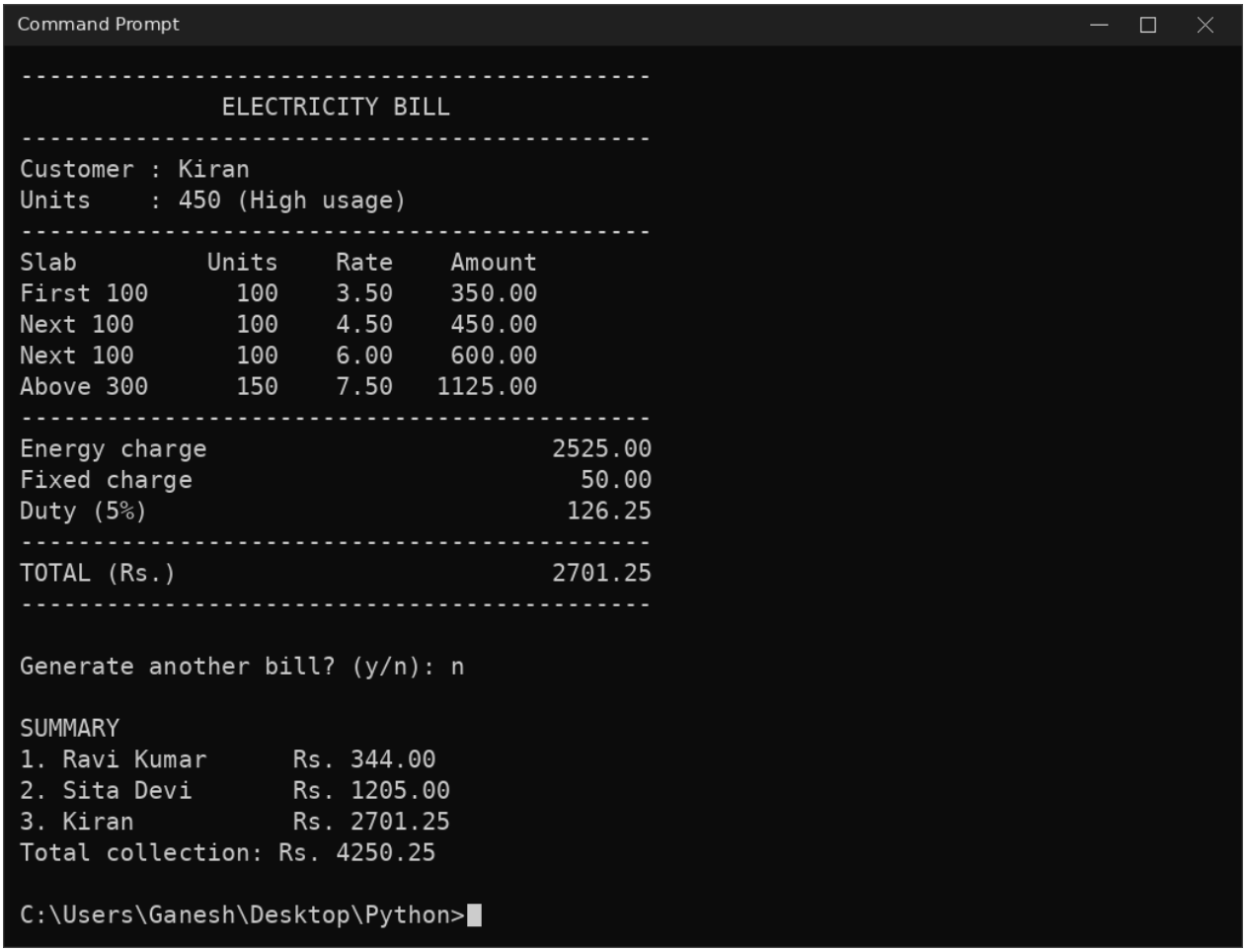
Program 8: Progressive Electricity Billing

File: progressive_electricity_billing.py

Output:



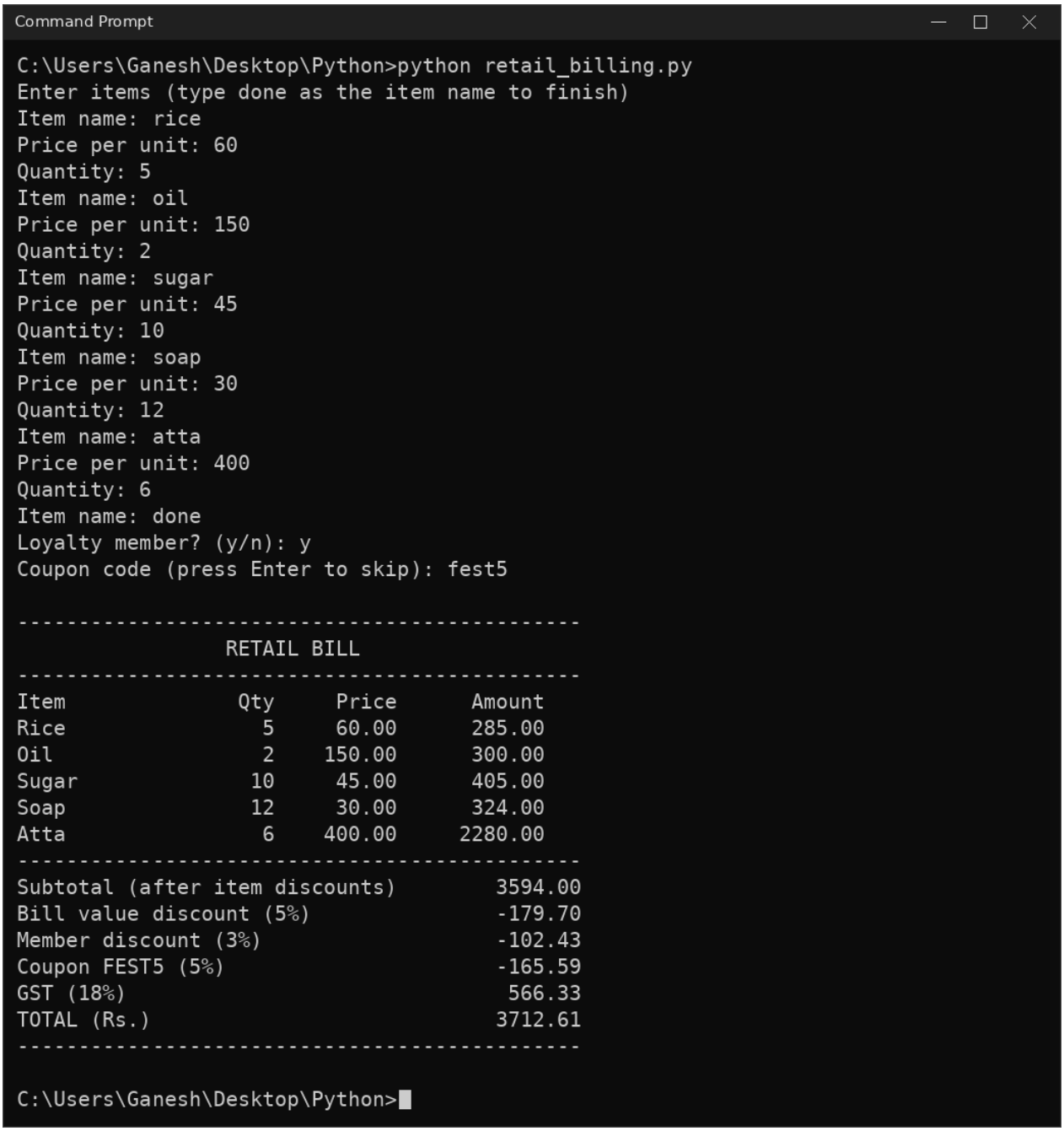

Command Prompt
Generate another bill? (y/n): y
Enter customer name: sita devi
Enter units consumed: 250
ELECTRICITY BILL
Customer : Sita Devi
Units : 250 (Moderate usage)
Slab Units Rate Amount
First 100 100 3.50 350.00
Next 100 100 4.50 450.00
Next 100 50 6.00 300.00
Energy charge 1100.00
Fixed charge 50.00
Duty (5%) 55.00
TOTAL (Rs.) 1205.00
Generate another bill? (y/n): y
Enter customer name: kiran
Enter units consumed: 450


Command Prompt
ELECTRICITY BILL
Customer : Kiran
Units : 450 (High usage)
Slab Units Rate Amount
First 100 100 3.50 350.00
Next 100 100 4.50 450.00
Next 100 100 6.00 600.00
Above 300 150 7.50 1125.00
Energy charge 2525.00
Fixed charge 50.00
Duty (5%) 126.25
TOTAL (Rs.) 2701.25
Generate another bill? (y/n): n
SUMMARY
1. Ravi Kumar Rs. 344.00
2. Sita Devi Rs. 1205.00
3. Kiran Rs. 2701.25
Total collection: Rs. 4250.25
C:\Users\Ganesh\Desktop\Python>

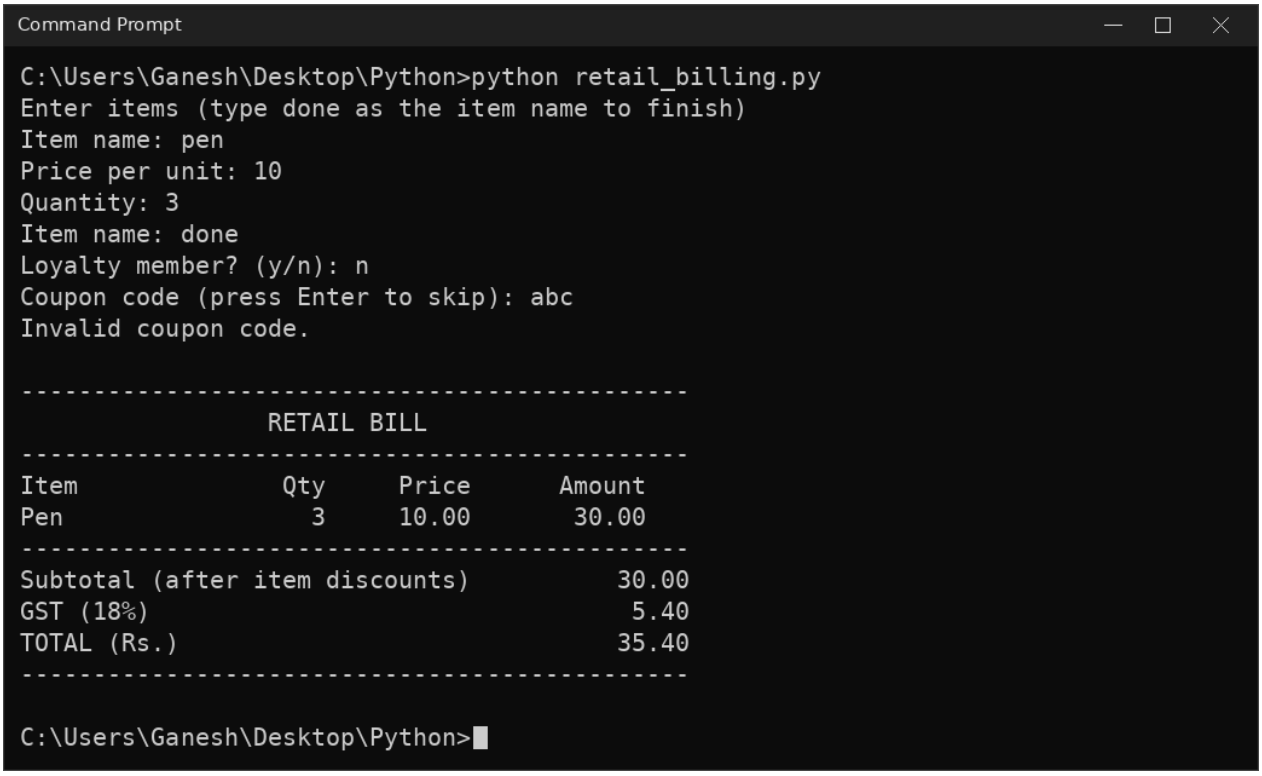
Program 9: Retail Billing

File: retail_billing.py

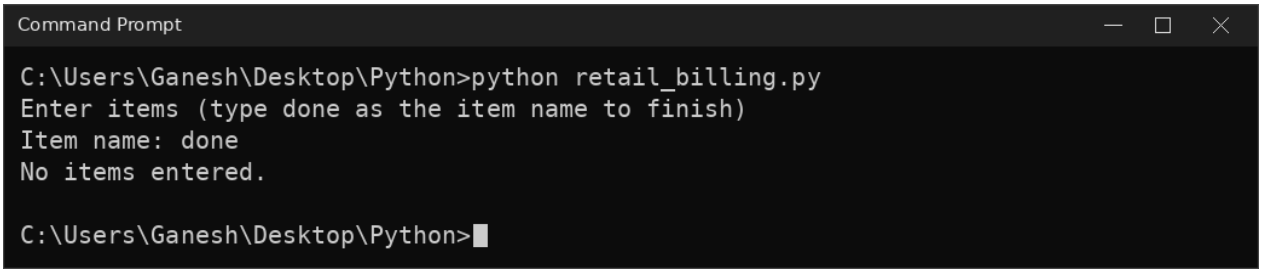
Output 1:



Output 2:



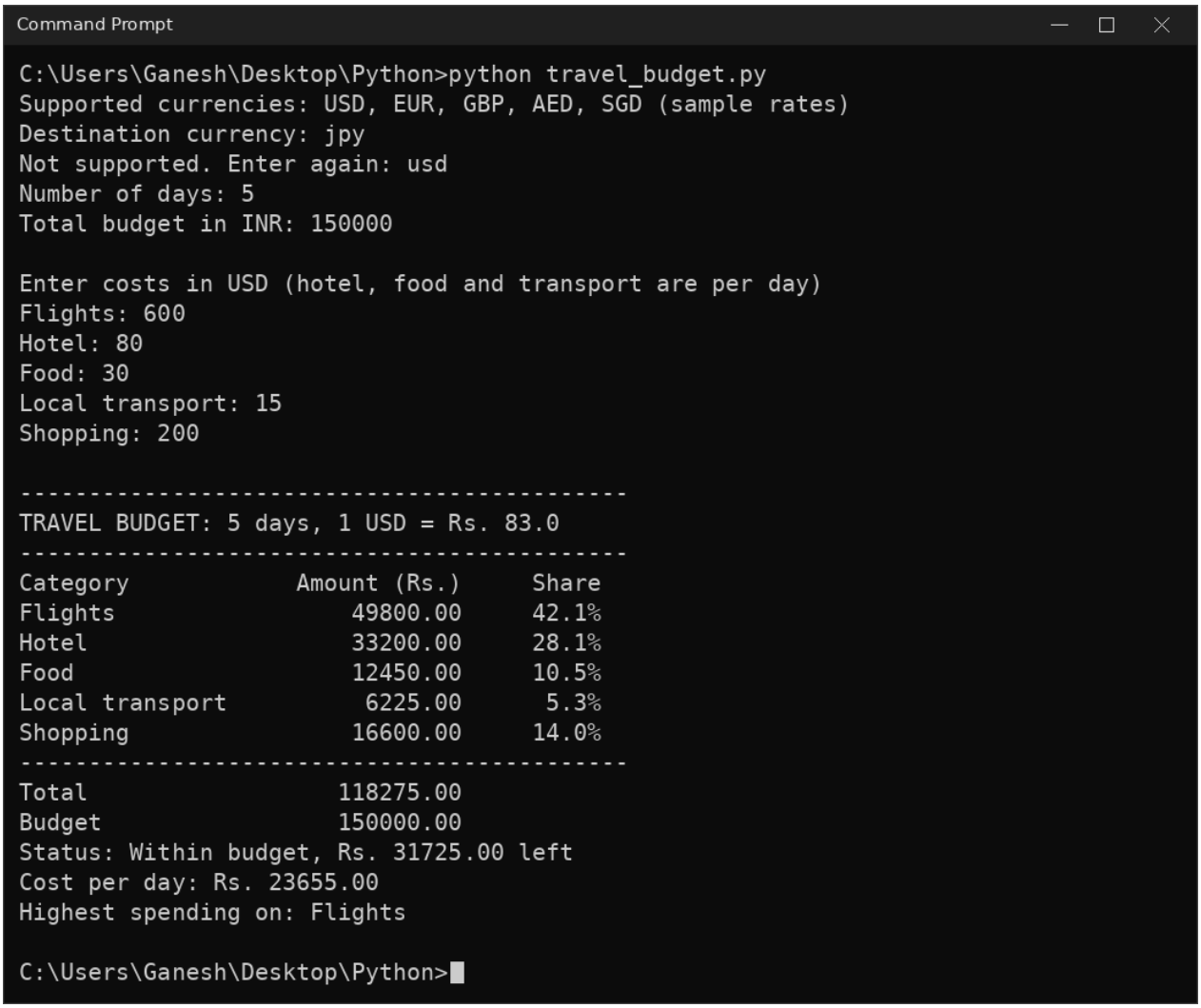
Output 3:



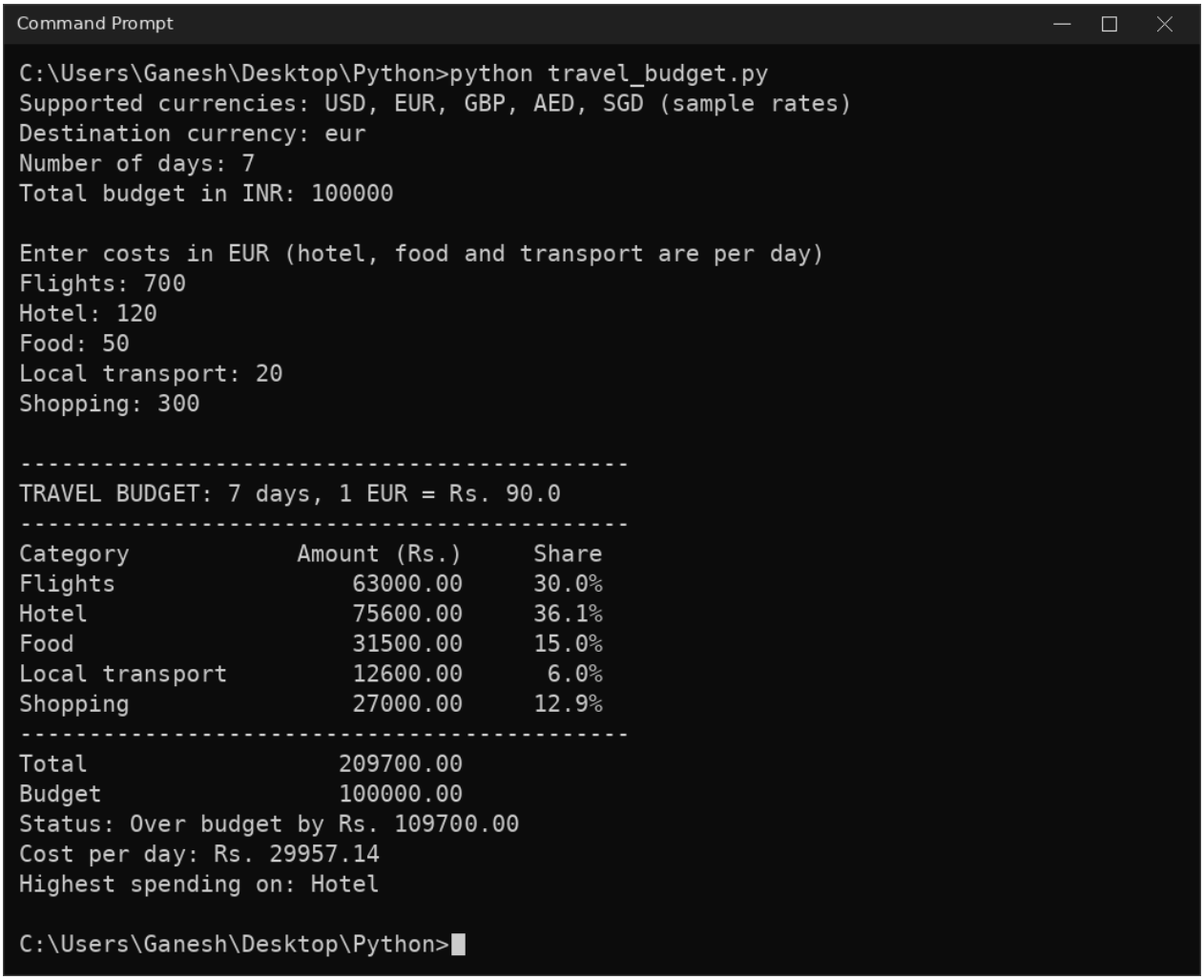
Program 10: Travel Budget Planner

File: travel_budget.py

Output 1:



Output 2:


Command Prompt
C:\Users\Ganesh\Desktop\Python>python travel_budget.py
Supported currencies: USD, EUR, GBP, AED, SGD (sample rates)
Destination currency: eur
Number of days: 7
Total budget in INR: 100000
Enter costs in EUR (hotel, food and transport are per day)
Flights: 700
Hotel: 120
Food: 50
Local transport: 20
Shopping: 300
TRAVEL BUDGET: 7 days, 1 EUR = Rs. 90.0
Category Amount (Rs.) Share
Flights 63000.00 30.0%
Hotel 75600.00 36.1%
Food 31500.00 15.0%
Local transport 12600.00 6.0%
Shopping 27000.00 12.9%
Total 209700.00
Budget 100000.00
Status: Over budget by Rs. 109700.00
Cost per day: Rs. 29957.14
Highest spending on: Hotel
C:\Users\Ganesh\Desktop\Python>